Password Cracker Simulator

Operating Systems (24CS2101)

Project
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Introduction /Problem Statement

This presentation explores a *password cracker simulation* designed to test password security using *brute-force attacks* and *hashing algorithms*. It highlights the simulation's practical application in understanding password vulnerabilities through multithreaded processing, aiming to enhance cybersecurity by demonstrating attack strategies and performance metrics.

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Password Cracker Simulation

Brute-force attack principles and hashing algorithms

The simulation uses brute-force to systematically generate all possible password combinations. It applies robust hashing algorithms to compare hashed inputs against targets, simulating realistic password cracking scenarios. This method reveals the critical role of hashing strength and complexity in protecting password integrity.

Multithreading and parallel processing concepts

To improve efficiency, the simulation distributes workload across multiple threads, enabling concurrent processing of password attempts. This parallelism shortens cracking time significantly, demonstrating key principles of multithreading for high-performance computing in cybersecurity applications.

Importance of strong passwords and secure hashing

Strong passwords combined with secure hashing algorithms form the first defense against unauthorized access. The simulation demonstrates how weak passwords can be cracked rapidly despite hashing. Employing complex passwords and robust hash functions significantly increases resistance to brute-force attacks, protecting sensitive data and maintaining system integrity.

Implementation and Performance Analysis

Algorithm design and keyspace division among threads

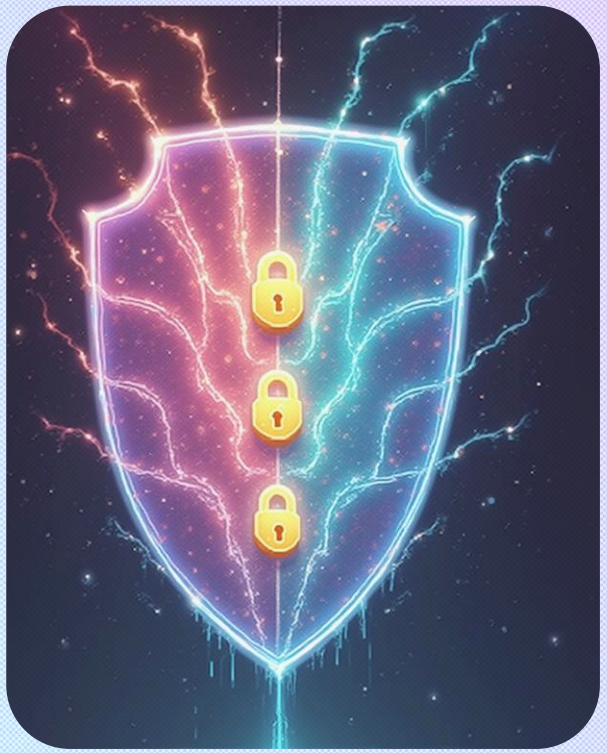
The brute-force algorithm is designed to systematically explore the entire password keyspace. To enhance efficiency, the keyspace is divided among multiple threads, allowing parallel search. This division ensures comprehensive coverage while minimizing overlap, optimizing resource utilization and accelerating the cracking process.

Thread synchronization and workload distribution

Effective synchronization among threads prevents conflicts and ensures balanced workload distribution. The simulation manages thread coordination to avoid redundant attempts, maintaining efficient use of CPU resources. Proper synchronization is crucial for stable and scalable parallel processing in computationally intensive tasks.

Performance metrics: attempts per second, execution time, CPU utilization

Performance is measured by attempts per second, total execution time, and CPU utilization. High attempts per second indicate efficient cracking speed. Reducing execution time while maintaining balanced CPU usage demonstrates the effectiveness of multithreading. These metrics guide optimization strategies for real-world password security testing.

Conclusions

The simulation highlights the critical role of *strong passwords*, *secure hashing*, and *efficient multithreaded algorithms* in cybersecurity. It showcases how parallel processing drastically improves cracking speed, emphasizing the need for robust defenses against brute-force attacks. This study reinforces best practices for password policies and system security enhancement.